

ALEX MORAN

SIMULATION | ROBOTICS | AUTONOMY | PLATFORM ENGINEERING

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PROFESSIONAL SUMMARY

Software engineer with 20 years of experience spanning autonomous systems, robotics, simulation infrastructure, scientific computing, and developer platforms. Built eVTOL autonomy R&D platforms, led advanced software teams, and architected multi-sensor robotic systems deployed to international nuclear-security organizations. Combines hands-on Python/C++ engineering with systems architecture, research-to-production execution, and cross-functional technical leadership.

CORE EXPERTISE

Autonomy & Simulation: Robotics, ROS/ROS2, Unreal Engine, Gazebo, ArduPilot, simulation environments, sensor fusion, SLAM, autonomous systems

Software & Scientific: Python, C++, C, JavaScript, SciPy, OpenCV, Cython, Node.js, API development, signal processing, machine learning

Platform & Data: Kubernetes, Linux/Ubuntu, platform engineering, TileDB, PostgreSQL, MySQL, Firebase, developer tooling

Technical Leadership: Software architecture, R&D platforms, SDK design, team leadership, field validation, research collaboration, technical productization

EXPERIENCE

Simulation Infrastructure Engineer | Wisk Aero

2025-Present

San Francisco Bay Area

- Developed and maintained a simplified vehicle first-order kinematics flight model in C++, integrating it into the full simulation loop.
- Developed and maintained C++ Vehicle Management System (VMS) models representing vehicle behavior and system interactions within the integrated simulation environment.
- Build simulation infrastructure supporting development and validation of autonomy capabilities for Wisk's self-flying eVTOL aircraft.
- Develop software and engineering workflows that make simulation environments repeatable, scalable, and effective for autonomy development.

Tech: C++, first-order kinematics, flight modeling, Vehicle Management System modeling, full-loop simulation

Lead Intelligent Systems Software Engineer | Supernal

2022-2024

Fremont, CA

- Built a research and development platform for eVTOL autonomy that was adopted by 12+ researchers across the R&D organization.
- Led the Advanced R&D software team and designed an SDK for vectorization, analysis, and reuse of sensor data across research workflows.
- Created simulation and developer tooling that enabled autonomy researchers to prototype and evaluate new capabilities in a shared environment.

Tech: Python, C++, JavaScript, ROS2, Unreal Engine, TileDB, SciPy, Kubernetes, Ubuntu

Scientific Engineer | Lawrence Berkeley National Laboratory

2017-2021

Berkeley, CA

- Architected a modular multi-sensor robotics platform supporting arbitrary combinations of physical sensors and sensor-fusion nodes; systems were used in field demonstrations and delivered to the Japanese Atomic Energy Agency, International Atomic Energy Agency, and U.S. laboratories.
- Developed human-robot interfaces for 3D gamma-ray sensing that enabled non-expert operators to interpret spectral and spatial data and make informed search decisions.
- Built autonomous UAV software for obstacle avoidance and higher-level behavioral decisions; validated the system in simulation and field testing.
- Contributed to PRISM and NG-LAMP, R&D 100 Award-winning radiation imaging and 3D mapping technologies, and to software later made available for licensing and commercialization.

Tech: Python, C, JavaScript, ROS, Unity, OpenCV, SciPy, Cython, Node.js, ArduPilot, behavior trees, Gazebo, AR/VR

EARLIER EXPERIENCE

Principal Engineer | AirGrub.com

2015-2016

Millbrae, CA

- Led software development for human and robotic food-delivery experiences at airports; client and concessionaire apps launched at multiple airports and robotic delivery was demonstrated in commercial environments.
- Designed product APIs and data systems spanning mobile applications, backend services, and operational workflows.

Tech: Objective-C, JavaScript, PostgreSQL, MySQL, Firebase, API development

Mobile Engineer | Khaylo.com

2014-2015

San Francisco, CA

- Built an iOS application for automatic exercise recognition using signal processing, machine learning, and sensor fusion; the system accurately tracked 20 exercise types.
- Co-invented technology for automatic recognition and management of human physical activity, later issued as U.S. Patent 10,022,071.

Tech: Objective-C, C, iOS, signal processing, machine learning, sensor fusion

DevOps Engineer | Eat24.com

2011-2014

San Bruno, CA

- Owned production infrastructure, security, monitoring, and developer tooling while supporting a rapidly growing online food-ordering platform.
- Helped develop the company's first native iOS application during a period of approximately 50% annual customer growth; Eat24 was later acquired by Yelp.

Tech: PHP, PHP-FPM, PostgreSQL, CentOS, Zabbix

Game Developer | Present Creative

2006-2011

San Francisco, CA

- Developed and shipped web and iOS games and interactive applications; published multiple titles before the company was acquired.

Tech: Objective-C, ActionScript, PHP, JavaScript

SELECTED RECOGNITION & PUBLICATIONS

- Contributor to R&D 100 Award-winning Berkeley Lab systems: Portable Radiation Imaging, Spectroscopy and Mapping (PRISM, 2019) and Neutron and Gamma Ray Source Localization and Mapping Platform (NG-LAMP, 2021).
- Named inventor, U.S. Patent 10,022,071 - "Automatic recognition, learning, monitoring, and management of human physical activities" (2018).
- Named investigator on Berkeley Lab software technologies for multimodal free-moving gamma-ray imaging and Radkit, a Python suite for light/ionizing-radiation data and robotics workflows.
- "Immersive Operation of a Semi-Autonomous Aerial Platform for Detecting and Mapping Radiation," IEEE Transactions on Nuclear Science (2021).
- "Improved Gamma-Ray Point Source Quantification in Three Dimensions by Modeling Attenuation in the Scene," arXiv:2104.11318 (2021).
- "3D Gamma-ray and Neutron Mapping in Real-Time with the Localization and Mapping Platform from Unmanned Aerial Systems and Man-Portable Configurations," arXiv:1908.06114 (2019).

EDUCATION

San Francisco State University | Bachelor's degree in Physics, 2004